Bio 203 Bio Tutorial

Associations Between Reef Disturbance, Habitat Characteristics, and Coral Species Richness Across Selected Philippine Marine Sites

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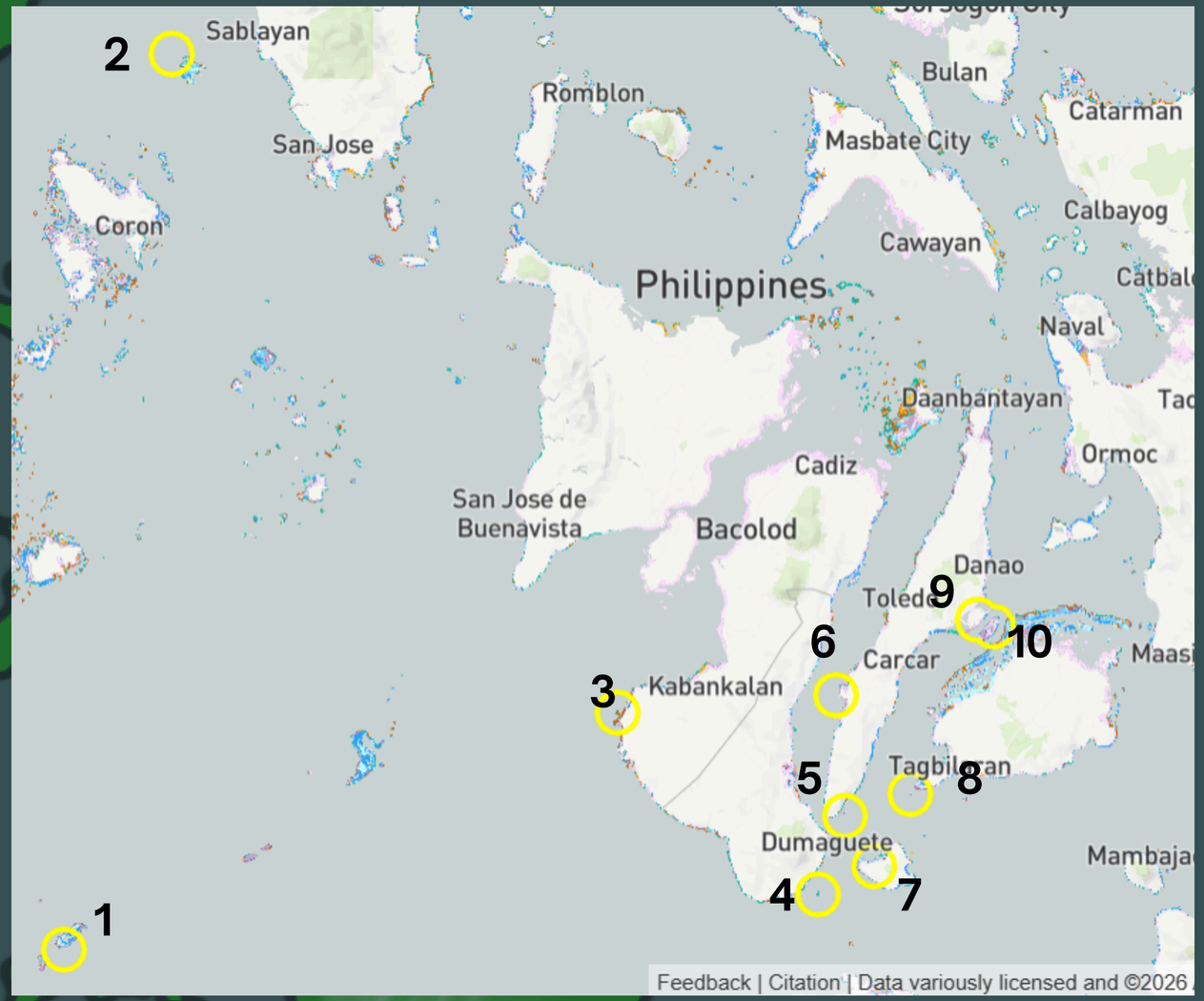
Introduction

- Coral reefs support diverse marine organisms and provide important habitats.
- Reef disturbance and habitat characteristics may influence marine biodiversity.
- Sea urchins are marine invertebrates that form part of reef communities.
- Species richness refers to the number of different species recorded in an area.
- This study compares 10 selected Philippine marine sites to explore possible associations among reef disturbance, habitat characteristics, and sea urchin species richness.

10 Chosen Sites

1. Tubbataha Reefs Natural Park
2. Apo Reef Natural Park
3. Danjugan Island
4. Apo Island, Negros Oriental
5. Sumilon Island, Cebu
6. Moalboal / Pescador Island
7. Siquijor Island reef area
8. Balicasag Island, Bohol
9. Olango Island reef area, Cebu
10. Mactan Island reef area, Cebu

Map of Chosen



Step 1 – Select and document the 10 sites

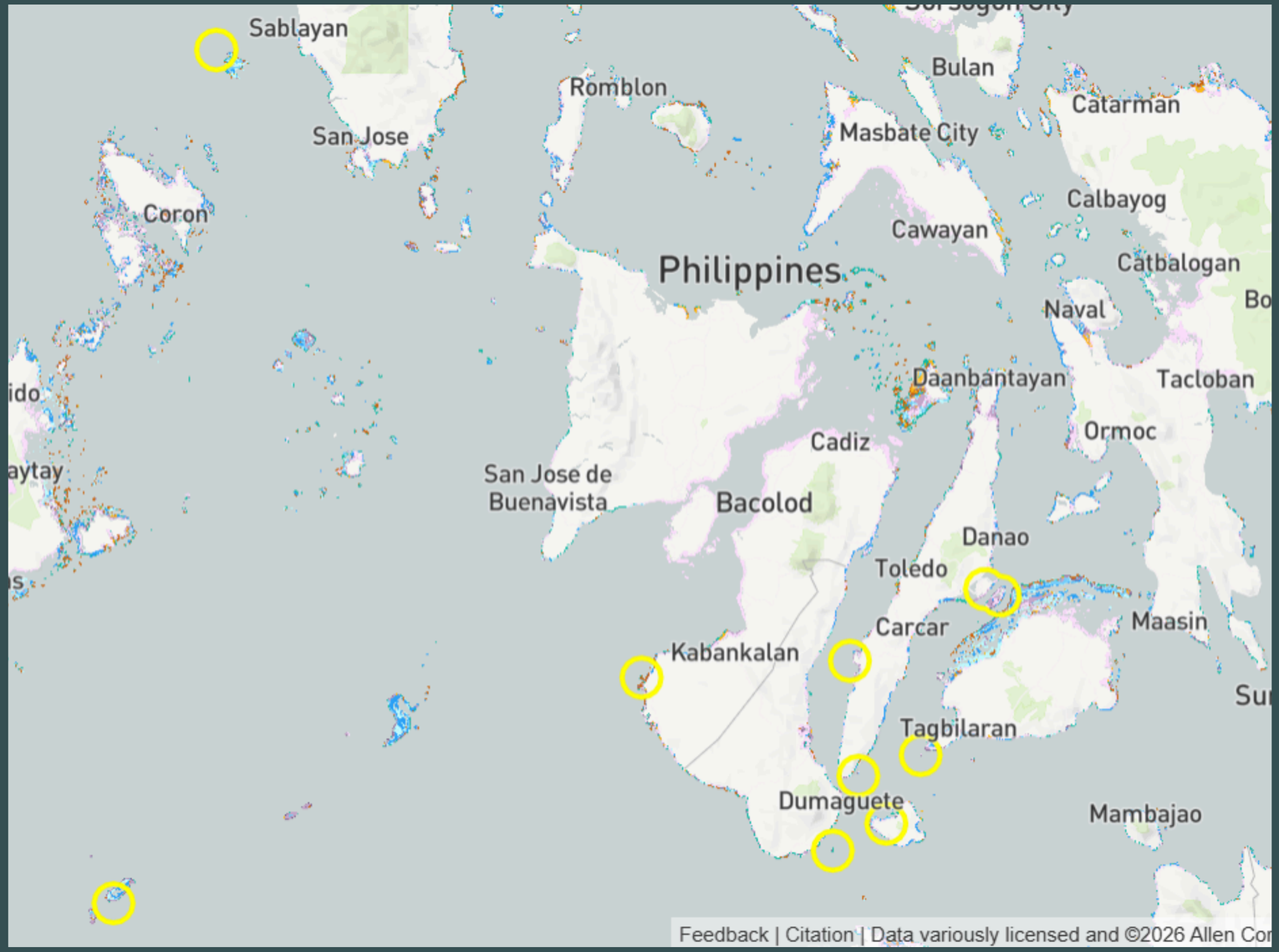
Site ID	Site Name	Latitude	Longitude	Disturbance Rank	Evidense or Source	Short Rationale
S01	Tubbataha Reefs Natural Park	8.8333	119.9167	1	UNESCO documents its remote location, protected status, and low fishing pressure. https://whc.unesco.org/en/list/653?utm	Candidate for very low local human disturbance.
S02	Apo Reef Natural Park	12.7464	120.3961	2	It is a designated marine protected area with extensive reef habitat. https://ctatlas.coraltriangleinitiative.org/MPA?id=4034000&utm	Candidate for low disturbance, although nearby human pressures still need consideration.

SO3	Danjugan Island	9.8729	122.3789	3	The surrounding reef is protected, and the island is managed as a conservation and ecotourism site https://doi.org/10.1002/aqc.3814Digital Object Identifier (DOI)	Candidate for relatively low disturbance, with tourism as a possible pressure.
SO4	Apo Island, Negros Oriental	9.0761	123.2702	4	The island has a community-managed marine sanctuary and a resident population. https://marineconservationphilippines.org/location/apo-island	Human influence is present, but conservation management may reduce certain pressures.
SO5	Sumilon Island, Cebu	9.42	123.39	5	Studies documented coral reefs and fishing pressure. https://academic.oup.com/icesjms/article/47/1/40/638514	Fishing activities may affect reef biodiversity and fish populations.
SO6	Moalboal / Pescador Island	9.95	123.35	6	Studies documented coral reefs, tourism, and fishing activities. https://doi.org/10.1016/j.biocon.2021.109242	Tourism and fishing may contribute to human pressure on reefs.

SO7	Siquijor Island reef area	9.20	123.52	7	Reef surveys reported varying coral cover and reef conditions. https://www.coast.ph/publications/2017-saving-philippines-reefs-report-siquijor-expedition	Differences in reef health make it useful for ecological comparison.
SO8	Balicasag Island, Bohol	9.5165	123.6802	8	The Coral Triangle Atlas documents a local fish sanctuary and no-take designation. https://ctatlas.coraltriangleinitiative.org/MPA?id=4007000	Protection is documented, but tourism and surrounding human pressures must also be considered.
SO9	Olango Island reef area, Cebu	10.25	124.05	9	Reports documented coral reefs, fishing, and coastal development pressures. https://www.keybiodiversityareas.org/en/site/factsheet/97	Human activities may affect coastal habitats and biodiversity.
SO10	Mactan Island reef area, Cebu	10.28	123.98	10	Coral surveys documented reef communities, and studies identified coastal development pressures. https://r7.emb.gov.ph/wp-content/uploads/2021/11/1st-Draft-EIS-2MLD-Crimson-Mactan-Desalination-Plant-Project-Nov-2021.pd	Coastal development may contribute to disturbance in nearby reef ecosystems.

Step 2 - Create the same 10-km buffer for Allen Coral Atlas and GBIF

Geomorphic



The map shows the selected coastal areas in the Philippines, marked by yellow circles. These locations are used to identify and observe the different reef formations through the Geomorphic layer of the Allen Coral Atlas. The map provides an overview of the distribution of the selected sites, which will be examined to compare their reef structures and geomorphic features.

Step 2 - Create the same 10-km buffer for Allen Coral Atlas and GBIF

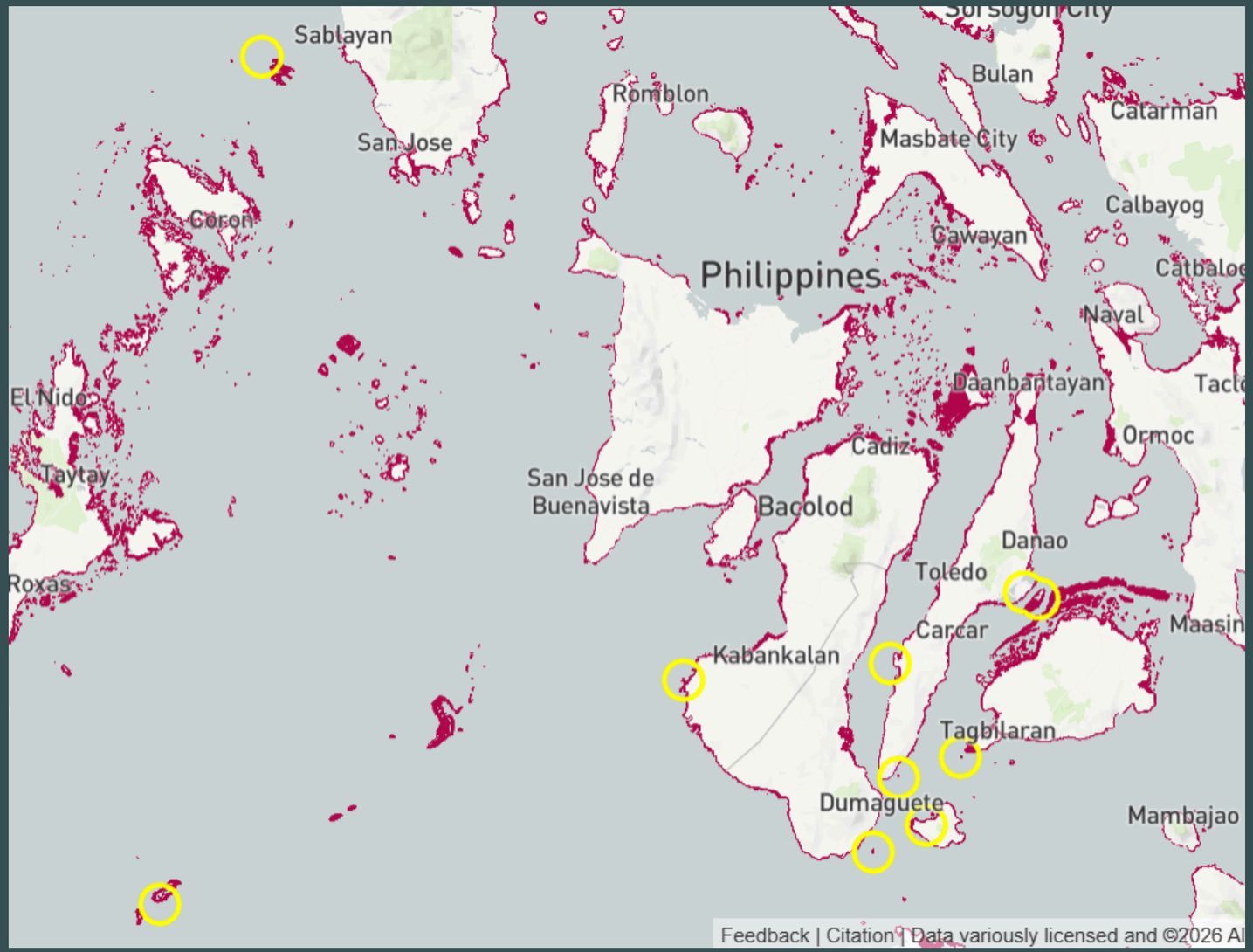
Benthic



The map shows the distribution of different benthic habitats in the selected coastal areas of the Philippines. The colored areas represent the mapped shallow reef habitats, which show that these habitats are mostly found along coastlines and around islands. The distribution also varies from one location to another, showing differences in the benthic composition of each study area.

Step 2 - Create the same 10-km buffer for Allen Coral Atlas and GBIF

Reef Extent



The map shows the mapped reef areas across the selected coastal locations in the Philippines. The pink markings indicate the extent and distribution of reef areas, which are mostly concentrated around islands and coastal regions. Some locations have more widely distributed reef areas, while others have smaller and more scattered reef formations.

Step 3 Allen Coral Atlas: describe reef topography (geomorphic zonation)

SITE ID	Description
Tubbataha Reefs Natural Park	This site is a slope and lagoon-dominated offshore atoll featuring balanced reef flats (45.95%) and steep back-reef slopes (24.19 %). It exhibits the highest reef crest development (2.72%) alongside deep lagoonal interiors (11.45%).
Apo Reef Natural Park	This site is a slope and lagoon-dominated oceanic atoll system featuring extensive back-reef slopes (29.66%) and deep/shallow lagoons (31.59% total). Outer flats and a distinct reef crest (2.56%) enclose the central lagoonal basin.
Danjugan Island	This site is a flat dominated fringing reef system characterized by broad outer (24.87%) and terrestrial (12.87%) reef flats. The seaward margin drops into a prominent sheltered reef slope (26.65%) and plateau zone.
Apo Island	This site is a strongly slope dominated oceanic island reef where sheltered slopes encompass over half (52.40%) of the mapped area. Narrow inner and outer reef flats fringe the steep volcanic shoreline with virtually no lagoon development
Sumilon Island	This site is a slope dominated fringing island system where sheltered (36.81%) and back-reef slopes (14.43%) form the primary habitat. Thin outer and terrestrial reef flats encircle the perimeter with minimal crest and lagoon features

Step 3 Allen Coral Atlas: describe reef topography (geomorphic zonation)

SITE ID	Description
Moalboal Pescador area	This site is a heavily flat dominated coastal shelf where inner (36.17%) and outer (33.14%) flats make up nearly 70% of the reef area. A thin sheltered slope (8.01%) borders the outer edge with almost non-existent lagoonal structure.
Siquijor reef area	This site is a flat-dominated fringing platform with broad outer (25.81%) and terrestrial (21.17%) reef flats. A well-defined sheltered slope (19.51%) and shallow lagoon pockets (7.00%) complete the platform structure
Balicasag Island	This site is a flat-dominated island platform dominated by expansive outer (38.67%) and terrestrial (27.96%) flats. The shallow flats drop sharply into narrow slopes (9.19% total) with minor shallow lagoon depressions.
Olango Island reef area	This site is a massive, flat-dominated shallow reef complex where outer (35.60%) and inner (28.72%) flats account for most of the expanse. Perimeter channels are formed by sheltered and back-reef slopes (16.71% combined) surrounding central tidal flats
Mactan Island reef area	This site is an overwhelmingly flat-dominated nearshore terrace with inner (37.50%) and outer (33.45%) flats covering over 70% of the total habitat. Slopes (7.25%) and lagoons (1.98%) are confined to narrow outer bands along the coast.

Step 4. Allen Coral Atlas: quantify benthic composition

Benthic class	Cover fraction	Habitat cover (%)	Area (km²)
Coral/Algae	0.2683	26.83%	48.3736
Seagrass	0.1013	10.13%	18.2591
Microalgal Mats	0.0124	1.24%	2.228
Rock	0.2048	20.48%	36.9202
Rubble	0.236	23.60%	42.5448
Sand	0.1772	17.72%	31.9546
Total	1	100.00%	180.2803

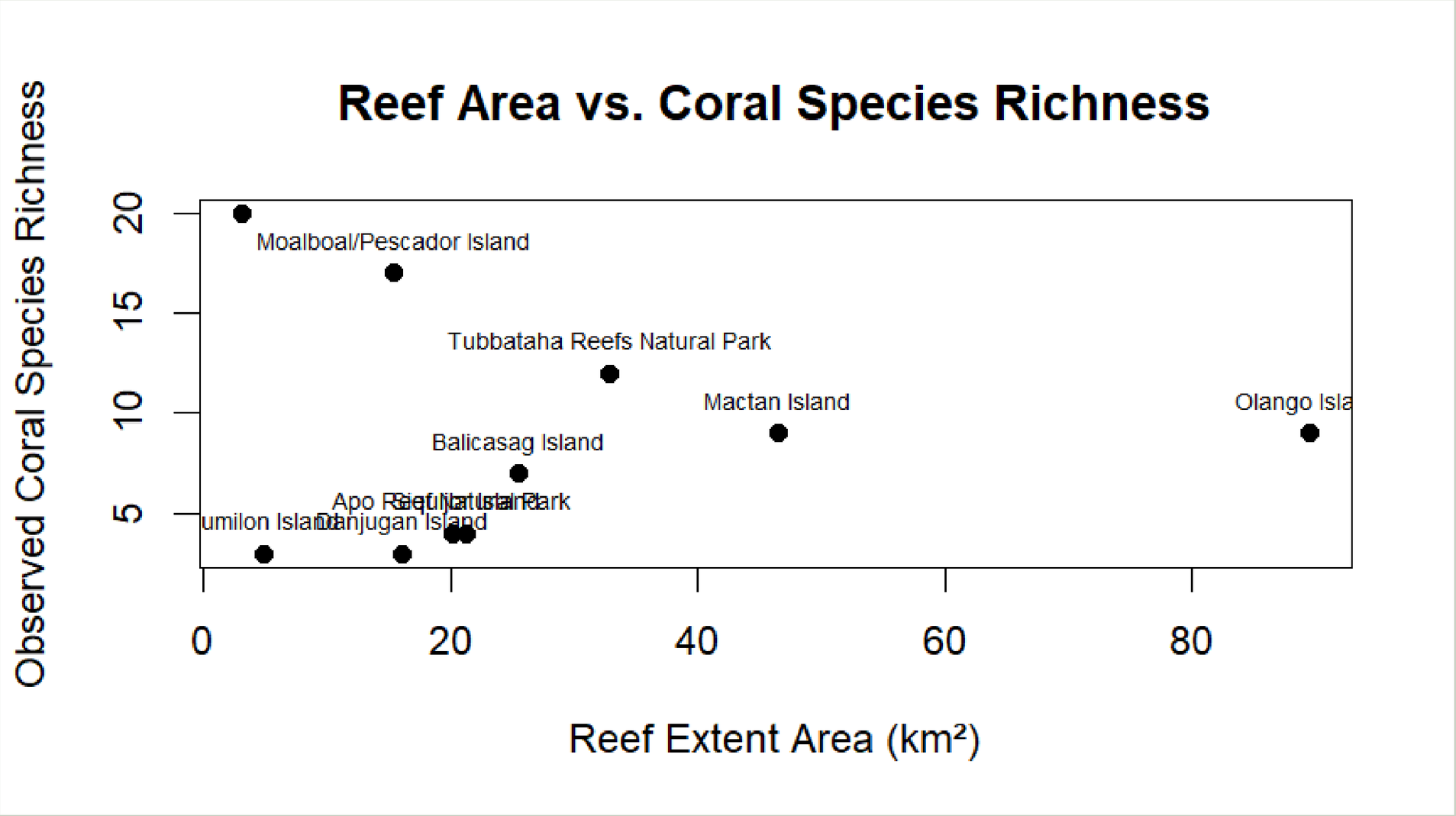
Step 5. Human Population Density and Disturbance

site_id	site_name	latitude	longitude	population_density_at_site_pixel	mean_population_density_within_20km_buffer	valid_raster_cells_in_buffer	rank_by_20km_mean_density
S01	Manjuyod coastal area	9.692	123.139	967.4	453.6	819	7
S02	Bais coastal area	9.591	123.122	1106.8	417.1	983	9
S03	Bindoy coastal area	9.746	123.139	946.9	443.7	818	8
S04	Amlan coastal area	9.462	123.215	734.3	640.9	989	5
S05	Sibulan coastal area	9.359	123.286	13297.7	799.5	813	3
S06	Dumaguete coastal area	9.307	123.305	10535.4	861.2	696	1
S07	Bacong coastal area	9.246	123.296	7222.0	830.8	660	2
S08	Dauin coastal area	9.191	123.265	2603.5	717.3	711	4
S09	Apo Island, Dauin	9.076	123.27	816.2	463.0	361	6
S10	Zamboanguita coastal area	9.104	123.199	2934.8	396.2	673	10

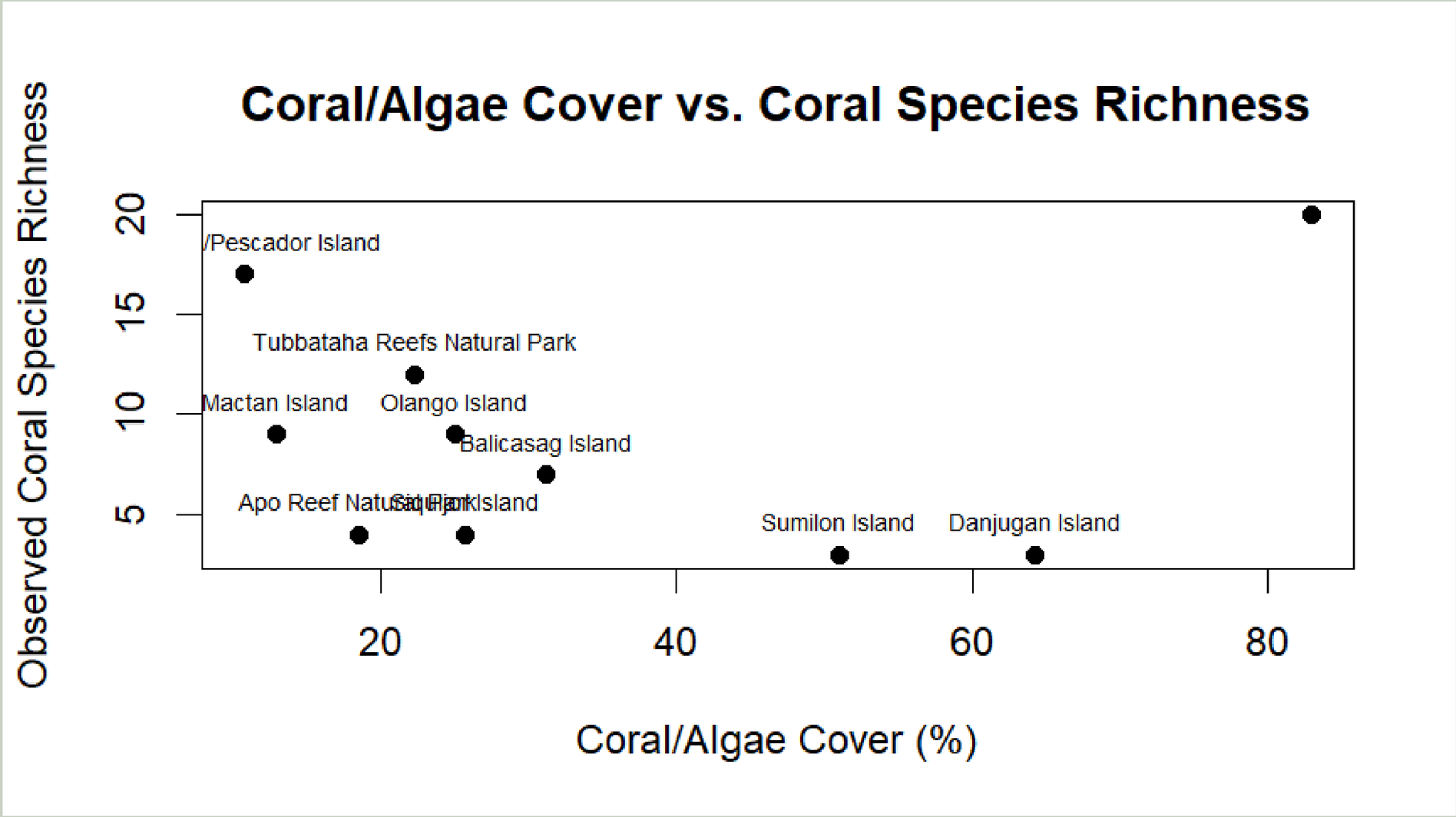
Step 6. GBIF multi-species extraction in R

Study site	GBIF records	Species richness
Tubbataha	41	12
Apo Reef	4	4
Danjugan	4	3
Apo Island	30	20
Sumilon	3	3
Moalboal/Pescador	25	17
Siquijor	5	4
Balicasag	12	7
Olango	15	9
Mactan	15	9

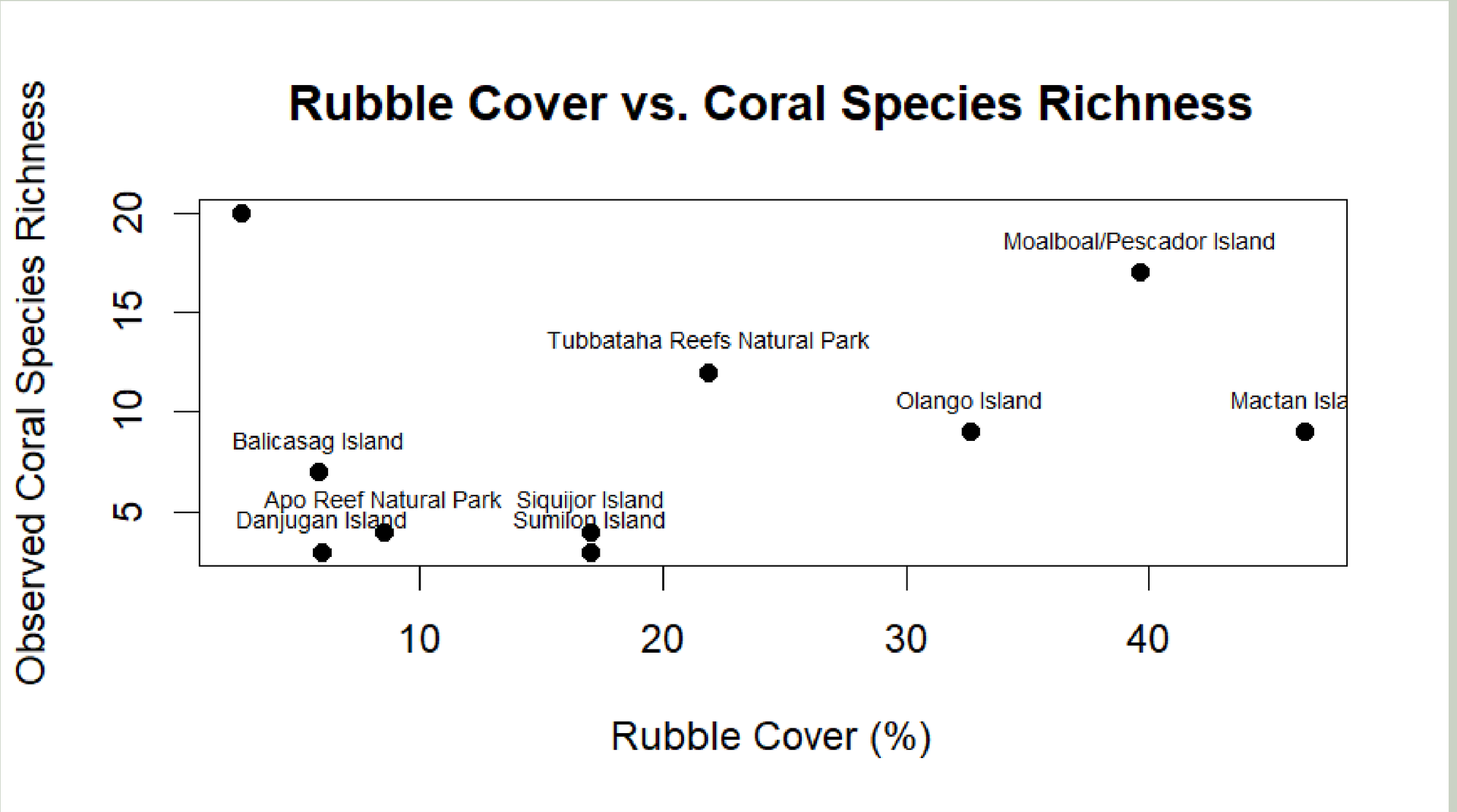
Step 7. Relationships Between Coral Species Richness and Environmental Factors



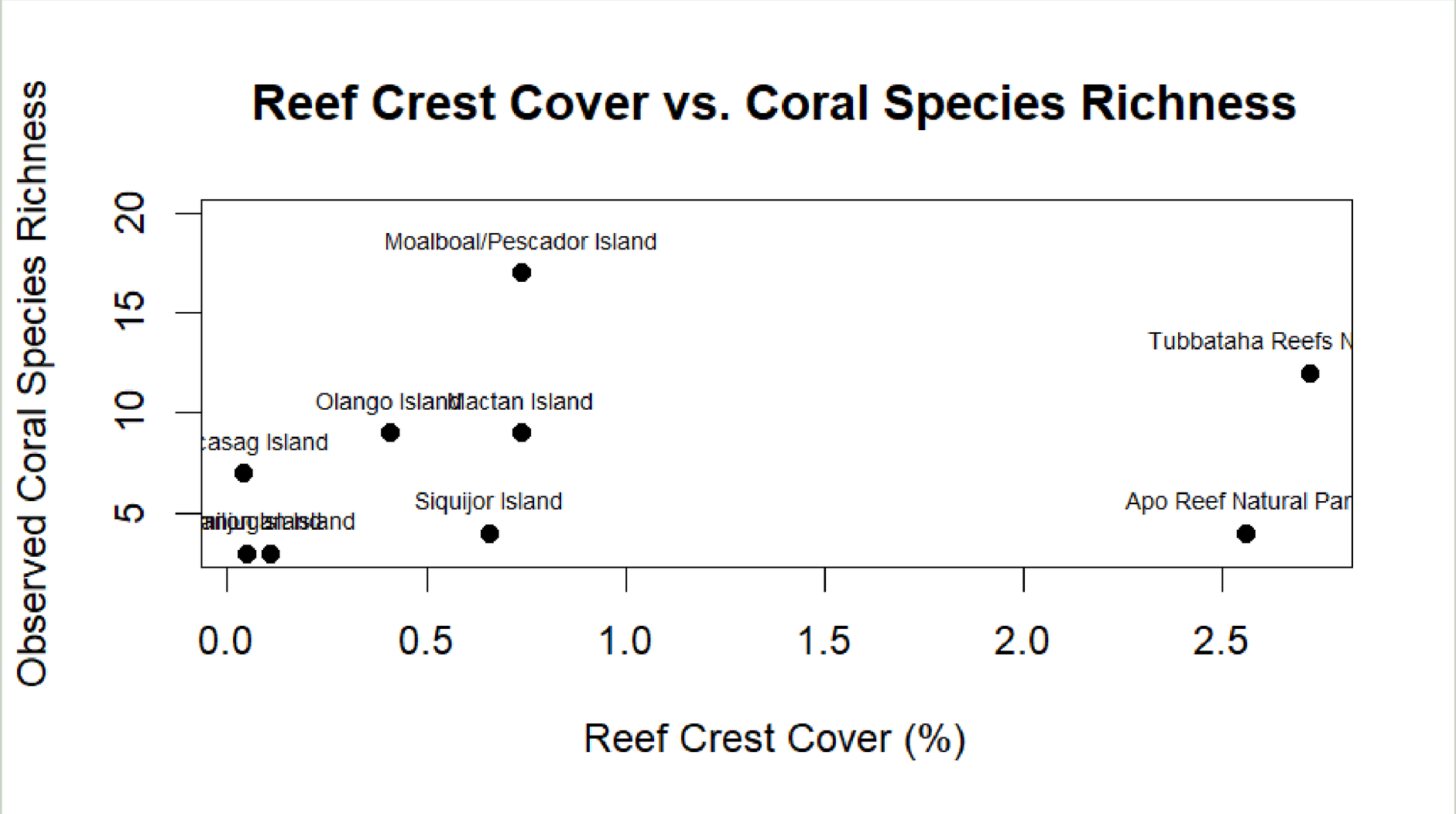
Step 7. Relationship Between Coral/Algae Cover and Observed Coral Species Richness Across the 10 Study Sites



Step 7. Relationship Between Rubble Cover and Observed Coral Species Richness Across the 10 Study Site



Step 7. Relationship Between Reef Crest Cover and Observed Coral Species Richness Across the 10 Study Sites.



Step 7. Spearman's Rank Correlation Between Environmental Variables and GBIF Recording Effort Across the 10 Study Sites

Environmental variable	ρ	p-value
Disturbance rank	0.006	0.9867
Reef extent area	0.213	0.5538
Coral/Algae cover	-0.268	0.4536
Rubble cover	0.232	0.5195
Reef Crest cover	0.519	0.1522
Outer Reef Flat cover	0.091	0.8016

Questions

1. What ecological disturbance gradient did your 10 sites represent and what evidence supports that gradient?
The sites form a gradient from very low to high disturbance. Tubbataha Reefs Natural Park and Apo Reef Natural Park are lowest because they are remote and well protected with low fishing pressure. Danjugan Island Apo Island and Sumilon Island show moderate disturbance from tourism and small nearby populations. Olango Island and Mactan Island rank highest due to dense coastal settlement fishing and coastal development. Population density data supports this pattern. Low disturbance sites had around 400 to 463 people per unit area within a 20 km buffer while high disturbance sites such as Dumaguete and Bacong exceeded 800 people in the same buffer zone.

2. How did reef topography differ among low and high disturbance sites?
Low disturbance sites such as Tubbataha and Apo Reef are oceanic atoll systems with well developed slopes at 24 to 30 percent distinct reef crests and extensive lagoons covering more than 31 percent of the area. Mid disturbance sites such as Apo Island and Danjugan Island are slope dominated fringing reefs with narrow flats and little to no lagoon development. High disturbance nearshore sites such as Olango Island and Mactan Island are overwhelmingly flat dominated. Reef flats make up nearly 70 percent of the habitat with very limited slopes and almost no lagoon or reef crest features.

3. Which benthic classes dominated each end of the gradient?
Across all sites combined coral and algae cover averaged 26.83 percent rubble made up 23.60 percent rock was 20.48 percent sand 17.72 percent seagrass 10.13 percent and microalgal mats 1.24 percent. Low disturbance sites tended to have higher coral and algae cover and greater structural complexity. High disturbance sites were dominated by reef flats rubble and sand with proportionally less living coral and more exposed or unstable substrates.

4. How did turbidity and nearby human population density vary among sites?
Turbidity was not measured directly but nearshore sites with high human density such as Olango Island and Mactan Island are expected to have higher turbidity from runoff sedimentation and construction. Population density showed a clear difference. Low disturbance sites had low surrounding densities ranging from 396 to 463 people in the 20 km buffer. High disturbance sites such as Dumaguete Bacong and Sibulan recorded 799 to 861 people in the same buffer zone. This reflects intense coastal settlement and associated pressures.

Questions

5. How many GBIF records and species were obtained at each site?

Tubbataha Reefs Natural Park had 41 GBIF records and 12 observed species. Apo Reef Natural Park had 4 GBIF records and 4 observed species. Danjugan Island had 4 GBIF records and 3 observed species. Apo Island had 30 GBIF records and 20 observed species. Sumilon Island had 3 GBIF records and 3 observed species. Moalboal and Pescador Island area had 25 GBIF records and 17 observed species. Siquijor Island reef area had 5 GBIF records and 4 observed species. Balicasag Island had 12 GBIF records and 7 observed species. Olango Island reef area had 15 GBIF records and 9 observed species. Mactan Island reef area also had 15 GBIF records and 9 observed species.

6. Were sites with high species richness also sites with high GBIF sampling effort?

Sites with higher GBIF sampling effort generally had higher observed species richness though the relationship was not perfectly consistent. Tubbataha Reefs Natural Park had the most records and relatively high species richness while Apo Island had the second highest number of records and the highest species richness overall. Moalboal and Pescador Island area also followed this pattern with many records and high species count. However some sites such as Apo Reef and Danjugan Island had very few records and correspondingly low species numbers which likely reflects limited sampling rather than true absence of species. Overall the data suggest that greater sampling effort tends to yield more observed species but differences in local habitat conditions may also influence the number of species recorded at each site.

7. Which environmental variable showed the clearest relationship with species richness for your taxonomic group?

No variable showed a statistically significant relationship. Rubble cover had a weak positive association with a Spearman rho value of 0.202. Disturbance rank had a very weak positive association at 0.080. Reef extent area showed a negligible positive association at 0.055. Coral and algae cover had a weak negative association at negative 0.294. All p values were well above 0.05 so none provided clear evidence of a meaningful link.

Questions

8. Did your simple GLM or regression support the group's ecological hypothesis? Describe direction and uncertainty not just the p value.

The hypothesis that greater disturbance or lower coral cover reduces species richness was not supported. Disturbance rank showed a slight positive trend meaning richness tended to increase marginally with higher disturbance. This trend was extremely weak and uncertain with wide scatter across sites. Coral and algae cover trended in the expected negative direction but the relationship was loose and inconsistent. Reef extent showed almost no directional trend at all. All relationships were highly uncertain with no consistent pattern emerging across the ten sites.

9. What is one plausible biological explanation for the pattern?

Coral species richness may depend more on local habitat complexity historical connectivity and consistent protection than on broad disturbance or cover measurements. Apo Island had moderate disturbance levels yet the highest richness likely due to long term community led management. This management maintains stable and diverse coral communities despite nearby human activity. Well enforced local protection can sustain richness even where regional pressure is moderate and this masks simple gradient based patterns.

10. What are at least three reasons the pattern might be biased or confounded?

First richness data came from unevenly surveyed sites so apparent differences may reflect how often an area was checked rather than true biological variation. Second disturbance correlates with island type distance from shore and larval connectivity. These natural biogeographic differences could drive richness independently of human impact. Third the study included only ten sites with narrow ranges in some variables. This makes it difficult to detect real relationships or to clearly separate human effects from natural environmental variation.

Limitations

One important spatial limitation is that Site 9 (Olango Island) and Site 10 (Mactan Island) are geographically adjacent with overlapping reef areas, so their habitat and disturbance data are not fully independent and may reduce the clarity of site-to-site comparisons. Species-richness records came only from GBIF, an opportunistic dataset rather than standardized synchronous field surveys, which creates uneven sampling and possible under-detection. The uniform 10-km buffer and broader coastal population-density estimates may not match each reef's actual human-influence boundaries. In addition, remote-sensing data from the Allen Coral Atlas captures large-scale features but misses small-scale, recent local disturbances that could affect coral richness.

Conclusion

This study assessed ten Philippine reef sites to examine how disturbance and habitat traits relate to coral species richness. Results show that well-developed reef structure, higher coral cover, and lower rubble cover correspond with greater species richness, whereas higher human population density and coastal development are linked to increased disturbance and reduced richness. Even protected sites face measurable human pressure, meaning formal protection alone is insufficient for reef health. Despite limitations—including overlapping reef areas for Olango and Mactan Islands, potential sampling bias, and broad-scale data—these findings reinforce that preserving natural reef habitats and managing local human pressures are essential for conserving coral biodiversity across the Philippines.

Borcelle

Thank You